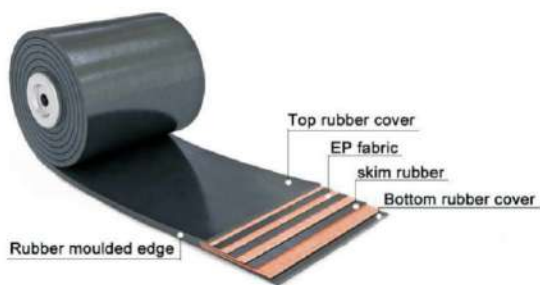




PRODUCT DESCRIPTION

To meet various needs from our customers all over the world, **SINOCONVE** mainly manufacture textile ply conveyor belting, facing industries like mining, crusher plant, cement, steel, construction and package handle throughout the world.

SINOCONVE multi-ply conveyor belting provide surpassing value for money to help our clients grow their business worldwide. Extensive producing technology and rich working experience enable us to manufacture high quality conveyor belts, excellently resisting to impact, abrasion, tear, cut, heat, fire, cold, oil and grease.



Property

- Suitable for heavy duty operations due to polyester/polyamid (EP) fabric structure design
- Low elongation with highly reliability and durability
- High adhesion between plies and between cover and ply.
- Advanced production technologies and facilities ensure ep conveyor beltings extended service life
- Available width from 300mm to 2500mm, depending on rubber conveyor belts thickness and length
- Different cover quality enable EP conveyor belts to transport materials which is -40℃ to 300℃, combustible, oily and cold.

SPECIFICATIONS



DIN-Y	1000m	EP400/3	4+2	9mm	300m	Moulded Edge
Cover Rubber Grade	Width	Tensile strength	Total Thickness	Length		Belt Edge type
			Top+Bottom Thickness			



SPECIFICATIONS

SINOCONVE can provide conventional rubber conveyor belts with the following specifications. If there are special requirements, we also provide customized services.

Cover Rubber Grade	8MPA,10MPA,12MPA,15MPA 18MPA,20MPA,24MPA,26MPA	DIN-X,Y,W RMA-1,RMA-2 N17,M24
Belt Width (Mm)	500,600,650,700,800,1000,1200 1400,1500,1800,2000,2200,2500	18",20",24",30",36",40",42" 48",60",72",78",86",94"
Tensile Strength	EP400/4,EP500/4,EP600/4 EP500/5,EP1000/5,EP1250/5 EP600/6,EP1200/6	330PIW, 440PIW
Top+Bottom Thickness	3+1.5, 4+2, 4+1.5, 4+3, 5+1.5	3/16"+1/16", 1/4"+1/16"
Belt Thickness	3mm,4mm,5mm,6mm,7mm,8mm,9mm,10mm,12mm,15mm,20mm,25mm	
Belt Length	10m, 20m,50m,100m,200m,250m,300m,500m	
Belt Edge Type	Moulded (Sealed) Edge OR Cut Edge	

FIRE RESISTANT CONVEYOR BELTS

For conveying combustible materials such as coal dust, gas, fertilizer, etc., it is very important to choose SINOCONVE refractory conveyor belt. It is mostly used in wood, paper and pulp, sugar and food, recycling and fertilizer plants.



SINOCONVE manufacture fire resistant conveyor belts according to customer's operational conditions.

Property

- Anti-static which conform to EN/ISO 20284 international standards
- When Ignited,it will self-extinguish within 15 seconds
- Rubber belts reinforced by several layers of textile fabrics (multi-ply) or steel cord
- Available in abrasion resistant type--as low as 150mm³,oil resistant type for both mineral oil and vegetable oil
- Widely used in underground mining industry for long distance transportation with high speed
- K grade(with covers) and S grade(with or without covers) are available for fire resistant conveyor belts



FIRE RESISTANT CONVEYOR BELTS

Flame retardant conveyor belts test based on EN/ISO 340:

- Anti-static: $\leq 3 \times 10^8$
- Drum friction: no flame under the condition of 325°C
- Buring test:



Flame retardant conveyor belt sample



Put six individual samples of the belt on a naked flame.



The flame goes out within 10 seconds after leaving the fire



Normal conveyor belt sample



Put six individual samples of the belt on a naked flame



Continue to burn after leaving the fire

- A. For individual test sample of fire resistant conveyor belts, the time it takes to self-extinguish should be no more than 15 seconds.
- B. For each group of six test pieces, the maximum cumulative time they take to self-extinguish should be no more than 45 seconds.

HEAT RESISTANT CONVEYOR BELTS



When conveying hot materials, the working surface rubber contact with materials directly. Conventional conveyor belts will be easily cracked and hardened under this condition. This will reduce the protection of carcass from cover rubber, thus resulting in carcass separation and splice failure.

SINOCONVE heat resistant conveyor belts select heat resistant compounds and innovative structure to solve these problems.

Ideal for transporting high-temperature, high-heat raw materials, semi-finished products, and products. If the belt surface temperature exceeds 60°C, use **SINOCONVE** heat-resistant conveyor belt.





HEAT RESISTANT CONVEYOR BELTS

Product advantage:

- Optimal formula of EPDM/POE vulcanizing,together with designed mixing process,to obtain ideal heat resistant cover rubber.
- Aminolysis reaction resistant fiber material is covered on both sides of textile carcass of burning through resistant conveyor belts.It not only effectively isolate polyester fiber from direct contact with cover rubber,preventing carcass destruction caused by heat,but also retain a high adhesion to EPDM under high temperature.
- Anti-aging property of special textile carcass make it no need to increase cover rubber thickness during belt design,resulting a reduction on cover rubber consumption and production cost of burning through resistant conveyor belts.

Item		Classes			
		T1	T2	T3	T4
		Test Temperature			
		≤100℃	≤125℃	≤150℃	≤175℃
		Change Range Allow			
Hardness	The difference after aging	20	20	20	20
	Maximum value after aging	85	85	85	85
Tensile Strength	Performance change rate	-25	-30	-40	-40
	Minimum value after aging	12	10	5	5
Elongation	Change rage after aging	-50	-50	-55	-55
	Minimum value after aging	200	200	180	180

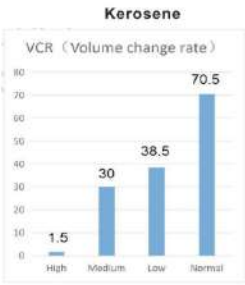
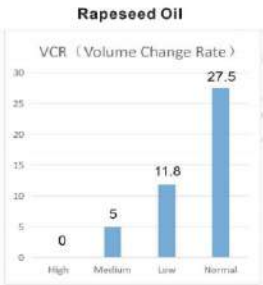
OIL RESISTANT CONVEYOR BELTS

SINOCONVE oil-resistant conveyor belt is very suitable for conveying oily materials in various fields such as food, animals, plants, and minerals. Using ordinary conveyor belts at high temperatures, due to the synergistic effect of oil and temperature, the life of the conveyor belts will be greatly shortened.



Oil Resistance Grade	Type	Typical Type
(1) Low	Cereals	Soybeans, Peanuts,Etc.
(2) Medium	Animal And Vegetable Oils	Rapeseed Oil, Palm Oil, Lard Oil, Fish Oil,Etc.
(3) High	Mineral Oil	Kerosene, Engine Oil, Hydraulic Oil, Etc.

Volume Change Rate / Type Of Belt





ACIDIALKALIRESISTANT CONVEYOR BELTS

SINOCONVE acid & alkali retardant conveyor belts is suitable for conveying acidic or alkaline materials in chemical plants, paper mills, cement plants, etc.

Typical Chemical Types That Can Be Applied

Chemical Type	Concentration	Chemical Type	Concentration
Hydrochloride	Less than 20%	Acetic Acid	Less than 10%
Nitric Acid	Less than 20%	Caustic Soda	100%
Sulfuric Acid	Less than 40%	Potassium Hydroxide	100%
Boric Acid	Less than 40%	Ammonia Anhydrous	100%
Phosphoric Acid	Less than 40%	Acetone	100%
Arsenic Acid	100%	Aniline	100%
Carbonic Acid	100%	Phenol	100%
Formic Acid	100%	Ethanol	100%

For other chemicals, please contact us.

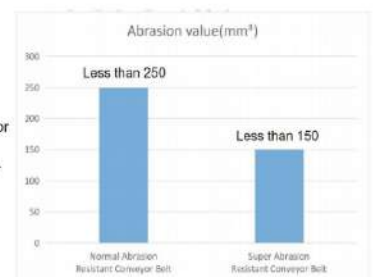
ABRASION RESISTANT CONVEYOR BELTS

SINOCONVE abrasion retardant conveyor belt uses rubber with excellent wear resistance, which is suitable for quarries, mines, etc.



SINOCONVE BELT Abrasion resistance test method: press a cylindrical sample with a diameter of 16mm and a thickness of about 8mm on a roller wrapped with sandpaper under a certain load. The sample wears when the drum rotates, and the wear after a certain distance is measured.

SINOCONVE Abrasion resistance test
① Normal abrasion resistant conveyor belt, the abrasion is less than 250mm³
② Super abrasion resistant conveyor belt with abrasion less than 150mm³





IMPACT/CUT RESISTANT CONVEYOR BELTS

SINOCONVE impact/cut retardant conveyor belts has excellent impact resistance, cut resistance and chipping resistance, and uses cover rubber. it is suitable for the transportation of sharp objects, such as crushers, rough stones, waste materials, logs, glass and iron flakes.

1. Impact test method

Testing machine: drop hammer impact testing machine

Cutting edge angle: 90°

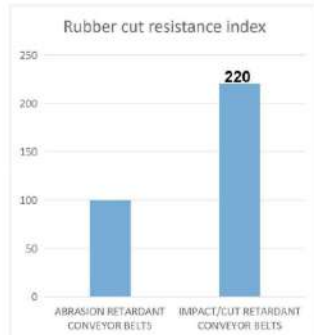
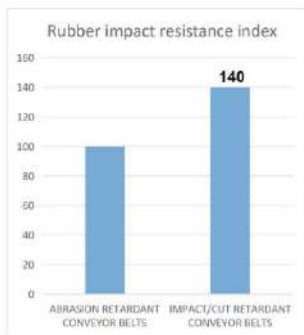
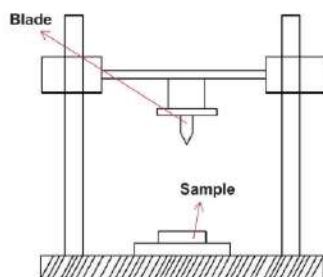
2. Cutting resistance test method

Testing machine: drop hammer impact testing machine

Drop weight: 15kg

Drop height: 400mm

Cutting edge angle: 90°



CHEVRON CONVEYOR BELTS

Chevron conveyor belts are used on slope angles up to approx. 30° for slightly rolling materials such as gravel and coal and up to approx. 40° for sticky materials such as wet sand and earth. Patterned conveyor belts are also a highly effective belt for conveying packages such as sacks and bales.

Product advantage of chevron conveyor belts:



1.The cleats are moulded and vulcanized in one single process together with the base belt of patterned conveyor belts.



2.Profile conveyor belts allows the use of smaller pulley diameters.



3.High quality cover rubber compound is excellent resistant to abrasion.



CHEVRON CONVEYOR BELTS

Chevron Conveyor Belt Drawing Data

Drawing	Cleat width mm	Cleat pitch mm	Cleat height mm	Belt width mm
	700-1500	5	1.5	700-1500
	460	50	3	500
	560		3	600
	660		3	700
	760		3	800
	860		3	900
	960		3	1000
	1160		3	1200---1600
	900	400	15	400-1000
	450	300	25	450-800

CHEVRON CONVEYOR BELTS

Drawing	Cleat width mm	Cleat pitch mm	Cleat height mm	Belt width mm
	620	330	32	700-850
	500	250	35	800
	500	250	35	600-900



CHEVRON CONVEYOR BELTS

Drawing	Cleat width mm	Cleat pitch mm	Cleat height mm	Belt width mm
	1450	250	35mm height 35mm width	400-1600
	500	500	50	600/750
	530	750	50	800
	300-1600	70	6	300-1600
	500-2400	75	6	500-2400

CHEVRON CONVEYOR BELTS

Drawing	Cleat width mm	Cleat pitch mm	Cleat height mm	Belt width mm
	14 inch	50	6	14 inch
	16 inch		6	16 inch
	18 inch		6	18 inch
	20 inch		6	20 inch
	24 inch		6	24 inch
	28 inch		6	28 inch
	32 inch		6	32 inch
	36 inch		6	36 inch
	40 inch		6	40 inch
	400-1200	80	10	400-1200
	400-1600	200	10	400-1600



CHEVRON CONVEYOR BELTS

Drawing	Cleat width mm	Cleat pitch mm	Cleat height mm	Belt width mm
	500	200	13.5/14	500
	600	200	13.5/14	600
	700	200	13.5/14	700
	800	200	13.5/14	800
	1000	200	13.5/14	1000
	550	250	15	600-1400
	750	300	15	800-1400
	440	250	15	650
	330	250	15	450
	540	250	15	750
	330	250	15	400-1400
	500	250	15	550-1400
	820	250	15	850-1400
	1200	250	15	1200-1400

CHEVRON CONVEYOR BELTS

Drawing	Cleat width mm	Cleat pitch mm	Cleat height mm	Belt width mm
	380	250	15	1400
	600	250	15	1400
	750	250	15	1400
	440	300	15	500-1600
	630	300	15	700-1600
	800	300	15	850-1600
	1000	300	15	1200-1600
	800	80	16.5	400-900
	800	80	16.5	400-900
	450	330	17	500-800
	630	330	17	650-1000
	800	330	17	900-1400
	950	330	17	1000-1600



CHEVRON CONVEYOR BELTS

Drawing	Cleat width mm	Cleat pitch mm	Cleat height mm	Belt width mm
	510	330	20	800
	1100	330	20	1600
	450	250	25	500-800
	550	250	25	650-800
	750	330	25	850-1400
	950	330	25	1400-1600
	1000	300	25	1400-2000
	230	250	25	300-400
	330	250	25	400-500
	440	250	25	500-650
	540	250	25	600-1000
	750	250	25	800-1000

CHEVRON CONVEYOR BELTS

Drawing	Cleat width mm	Cleat pitch mm	Cleat height mm	Belt width mm
	750	350	25	900-1400
	550	250	25	600-1200
	1000	400	25	1200-1400
	580	330	32	800
	750	330	32	1600
	5			
	800	80	16.5	400-900
	800	330	32	900-1100



SIDEWALL CONVEYOR BELTS

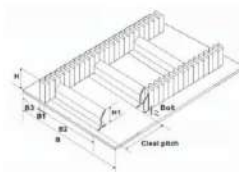


Sidewall conveyor belts are composed of three parts: base belt, sidewall and cleats
Possible to convey at steep angle with big capacity in a limited space.



Special treated steel mesh inserted into both top cover and bottom cover, creating an excellent combination of longitudinal flexibility and transverse rigidity.
The entire steep angle conveyor belts are highly durable, low elongation and long lasting

SINOCONVE Sidewall conveyor belts are composed of three parts: base belt, sidewall and cleats. Sidewall and cleats are connected with base belt by second hot vulcanizing, which create a super-strength bond between sidewall, cleats and base belt, preventing base belt aging caused by multivulcanizing. The entire steep angle conveyor belts are highly durable, low elongation and long lasting.



Sidewall is hot vulcanized on the base belt
High quality rubber to guarantee sidewall withstand highly stress and repeated flexing.



Fabric inside sidewall perfectly combine flexibility and strength to effectively avoid torn.



Cleats are also hot vulcanized on the base belt

Fabric inside cleats protect cleats from deformation caused by impact.



Cleats are the main part to carry materials in a high capacity and steep angle

Available with TC or TCS type

Special compounds to extremely avoid deformation caused by impact.

Base belt width(B)	Sidewall height(H)	Cleat height(H1)	Bottom width of sidewall(B1)	Cleat width(B2)	Empty width(B3)
300	40	35	25	180	35
	60	55	50	120	40
	80	75			
400	60	55	50	180	60
	80	75			
	100	90			
500	80	75	50	250	75
	100	90			
	100	90			
650	120	110	50	350	100
	160	140	75	300	
	120	110	50	460	
800	160	140	75	410	120
	200	180			
	200	180			
1000	200	180	75	550	150
	240	220			
	160	140			
1200	200	180	75	690	180
	240	220			
	300	260			
1400	200	180	75	830	210
	240	220	100	780	
	300	260			
	400	360			



SIDEWALL CONVEYOR BELTS



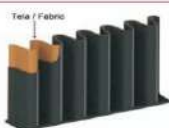
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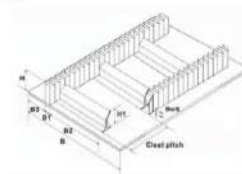


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300	40	35	25	180	35
	60	55	50	120	40
	80	75		180	
400	60	55		50	250
	80	75			
	100	90			
500	80	75	50	250	75
	100	90			
	120	110			
650	160	140	75	300	100
	120	110	50	350	
	100	90	50	460	
800	160	140	75	410	120
	200	180			
	240	220			
1000	160	140	75	550	150
	200	180			
	240	220			
1200	200	180	75	690	180
	240	220			
	300	260			
1400	200	180	75	830	210
	240	220			
	300	260			
	400	360	100	780	



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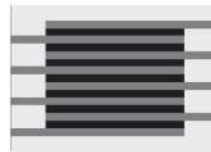
Property

- Suitable for heavy duty operations due to polyester/polyamid(EP) /steel cord fabric / steel cord structure design.
- Low elongation with highly reliability and durability.
- High adhesion between plies and between cover and ply.
- Advanced production technologies and facilities ensure ep conveyor beltings extended service life
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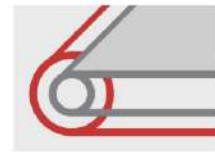
STEEL CORD CONVEYOR BELTS

Based on more than 20 years of experience in development, manufacturing and application know-how, **SINOCONVE** Conveyor Belts meet specific end-user requirements for cost efficiency and reliability. With in the **SINOCONVE** product range, steelcord belts are the ideal choice for long, high-tension or heavy-duty installations.

Advantages of Steelcord Conveyor Belts



Enhanced safety and service life due to high splice strength.



Small pulley diameters due to high splice strength reduces the size of conveyor components, particularly gearboxes.



Higher impact resistance, good for heavy-duty applications.



Increased conveyor lengths lower expenditure on plant. Longer belt life, fewer transfer points, less pollution.



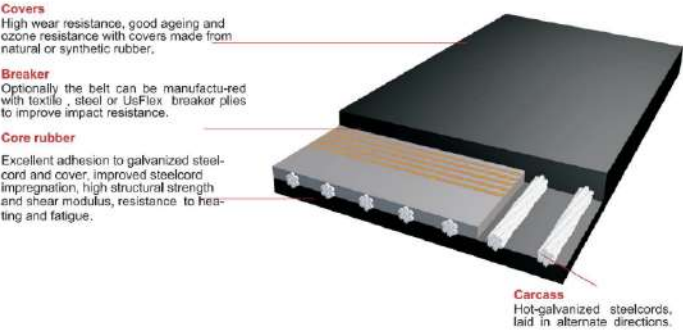
Increased troughability, improved conveying capacity maintaining the same belt width.



Lower elongation, smaller take-up, equipment.



Belt construction



Steel cord and core rubber

The bond between the core rubber and the metallic surfaces as well as the complete filling of the spaces between the wires are essential for the long-term integrity of the belt carcass.

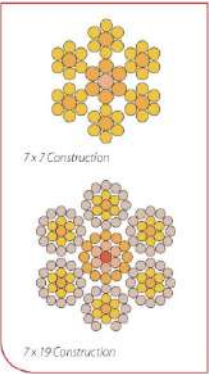
The quality of this bond is constantly observed by regularly performing a series of static and dynamic laboratory tests, such as adhesion, pull-out strength, differential pressure test or torsion-bending test.

Standard cords of SINOCONVE Steelcord Belts have an elongation of between 0.22 and 0.28%. Testing is carried out dynamically with loads between 2% and 20% nominal belt rating.

Special steelcord designs have been developed. The standard hot-galvanized cord types are 7x7 and 7x19 and range from 3.0 to 13.0 mm diameter. The open cross-lay design ensures that the rubber penetrates thoroughly.

The chemical adhesive system achieves superior bonding to the steelcord. This bond, together with the open cord construction ensures a high pull-out strength between steelcord and core rubber as well as between center and outer strand.

Cover damages may result in the entry of moisture to the steelcord layer. To achieve optimum corrosion protection and long



PRODUCT RANGE

The SINOCONVE standard range is shown in the table. It is based on DIN 22131 as well as on European standard EN ISO 15236, type A1. SINOCONVE can supply most existing belt types following different. Please contact us for enquiries about belt types according to the EN types A2, for belts wider than 2400 mm, or other types that differ from SINOCONVE's standard.

STEEL CORD CONVEYOR BELT CONVEYOR MODEL SERIES TABLE		500	630	800	1000	1250	1400	1600	1800	2000	2250	2500	2800	3150	3500	4000	4500	5000	5400	6300	7000	7500
Model		500	630	800	1000	1250	1400	1600	1800	2000	2250	2500	2800	3150	3500	4000	4500	5000	5400	6300	7000	7500
Min breaking strength F _{min} (kN)		500	630	800	1000	1250	1400	1600	1800	2000	2250	2500	2800	3150	3500	4000	4500	5000	5400	6300	7000	7500
Wear tear resistance index I _W (mm)		3.0	3.0	3.5	4.0	4.5	5.0	5.0	5.0	5.0	5.0	5.0	5.0	5.0	5.0	5.0	5.0	5.0	5.0	5.0	5.0	5.0
Min breaking force of wire rope F _{min} (kN)		7.8	7.0	8.9	12.9	16.1	20.6	25.5	25.5	28.2	40.0	39.6	50.5	56.0	63.5	76.3	91.0	96.2	130.4	142.4	168.7	
Steel cord spacing (mm)		14.0	10.0	10.0	12.0	12.0	14.0	12.0	13.5	12.0	11.0	15.0	13.5	15.0	15.0	15.0	16.0	17.0	17.0	19.5	19.5	21.0
Min cover thickness t _{min} (mm)		4.0	4.0	4.0	4.0	4.0	4.0	4.0	4.0	4.0	5.0	5.0	5.0	5.0	5.0	6.5	7.0	7.5	8.0	10.0	10.0	10.0
Steel cord Ø (mm)	Limit deviation	Number of steel cord n																				
500	+10/-6	33	45	45	39	39	34	39	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
630	+10/-7	44	60	60	51	51	45	51	46	52	56	41	46	41	41	39	36	N/A	N/A	N/A	N/A	N/A
800	+10/-8	54	75	75	63	63	55	63	57	63	69	50	57	50	50	51	48	45	45	N/A	N/A	N/A
1000	±10	66	95	95	79	79	63	79	71	79	86	64	71	64	64	59	55	55	N/A	N/A	N/A	N/A
1250	±10	83	113	113	94	94	82	94	85	94	104	76	85	76	77	77	71	66	66	59	59	54
1400	±12	96	133	133	111	111	97	111	130	111	122	88	99	89	90	90	84	78	78	69	69	64
1600	±12	111	151	151	126	126	111	126	114	126	140	101	114	101	104	104	96	90	90	78	80	73
1800	±14	125	171	171	143	143	125	143	129	143	159	114	128	114	117	109	102	102	89	90	83	
2000	±14	139	191	191	159	159	139	159	144	159	177	128	143	128	130	130	121	113	113	99	100	92
2250	±15	153	211	211	176	176	154	176	159	176	195	141	158	141	144	144	134	125	125	109	110	102
2400	±15	167	231	231	193	193	166	193	174	193	213	155	173	155	157	157	146	137	137	119	119	110
2600	±15	181	251	251	209	209	182	209	189	209	231	168	186	168	170	170	159	149	149	129	129	120
2800	±15	196	271	271	226	226	197	226	203	226	249	181	202	181	183	183	171	161	161	139	139	129
3000	±15	210	291	291	243	243	211	243	216	243	268	195	217	195	195	195	183	172	172	149	149	139
3200	±15	224	311	311	260	260	225	260	233	260	286	208	232	208	208	208	196	184	184	160	160	149

Note: N/A Not applicable due to troughing



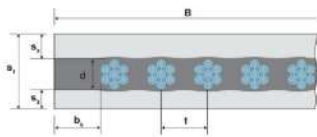
ADHESION OF STEEL CORD

Model	550	630	800	1000	1250	1400	1600	1800	2000	2200	2500	2800	3150	3500	4000	4500	5000	5400	6300	7000	7500
Before aging %	80.0	80.0	67.5	75.0	82.5	90.0	90.0	99.0	105.0	99.0	123.0	123.0	136.5	144.0	148.5	160.5	178.5	184.5	207.0	217.5	240.0
After aging %	50.0	50.0	57.5	65.0	72.5	80.0	80.0	89.0	95.0	89.0	113.0	113.0	126.5	134.0	138.5	150.5	168.5	174.5	197.0	207.5	230.0

Recommended diameter of ST belt pulley

Model	ST630	ST800	ST1000	ST1250	ST1600	ST2000	ST2500	ST3150	ST3500	ST4000	ST4500	ST5000	ST5400	ST6300
Minimum drum diameter(mm)	500	500	630	800	1000	100	1250	1400	1600	1600	1600	1800	1800	1800

- B belt width in mm
- d cord diameter in mm
- t cord pitch in mm
- s₁ belt thickness in mm (equivalent to s₂+d+s₃)
- s₂ top cover thickness in mm
- s₃ bottom cover thickness in mm
- b₀ solid rubber edge



Cover grades

The most important factor for the selection of the type of rubber quality is the nature of the conveyed material: lump size, shape, hardness, density, temperature or oil/chemical content. Operating conditions, such as ambient temperatures, radiation or safety requirements also play a role in choosing the correct cover.

SINOCONVE supplies rubber grades according to a wide range of international standards. Some standard cover grades are listed in the table below. All qualities are resistant against ozone, UV radiation, ageing and are antistatic according to ISO 284. Please consult us for special requirements.

COVER RUBBER PROPERTY

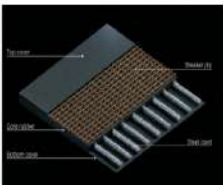
Test items		Property indexes					
		GB9770-2013			DIN22131-1994		
		D	H	L	W	X	Y
Tensile strength	mpa±	18	24	15	18	25	20
After aging(70℃,7d)	%±	400	450	350	400	450	400
Change rate of tensile strength	%	-25→+25	-25→+25	-25→+25	-25→+25	-25→+25	-25→+25
Change rate of elongation							
Abrasion mm ³	s	100	90	100	90	120	150

Note: D—strong wear H—strong scratches L—general working conditions

SINOCONVE grade	Tensile strength min. N/mm ²	Elongation at break min. %	Abrasion max. mm ³	Characteristics and Application
Flame resistant				
K	20	400	200	Abrasion resistant cover with flame resistant characteristics according to ISO 340
VF	17	350	175	Cover with fire resistant properties according to DIN, vt
Heat resistant				
T120	15	400	175	Heat resistant, abrasion resistant cover for temperatures up to 120°C



Breaker and rip detection



Steelcord belt with breaker ply

For applications where increased resistance against belt damage is required, SINOCONVE belts can be equipped with more effective protection. Additional transverse reinforcements (breakers) offer higher resistance to impacts or cuts and can be integrated in the top-, bottom- or both covers. Textile breakers consist of elastic polyamide cords with good capacity to absorb impact energy. More transverse strength can be achieved with a breaker consisting of elastic steel cords. The open construction of breaker plies guarantees maximum adhesion between the rubber layers. Even though the usage of breakers can prevent objects from ripping the belt, they cannot provide complete protection. A rip-detection system can be the choice for high-risk applications.

The system consists of sensor loops, vulcanized onto the belt, and a detector that monitors the condition of the loops. Severe damage to the belt will cause one or more sensor loops to rupture. This condition is detected and can trigger an emergency stop.

Type of breaker



Nylon fabric breaker ply

Breakers generally fall into two different categories or types. These are 'fabric breakers' (also referred to as textile breakers) and 'steel breakers'. A wide range of textile fabrics (mostly nylon) in various strengths and densities are used to make fabric breakers. Lighter-weight fabric versions (polyester/nylon) are designed to simply absorb and dissipate energy while stronger, heavier weight nylon breakers and steel breakers can actually stop the belt, which limits the amount of damage even more effectively. These are often referred to as 'Rip Stop' breakers.



Steel breaker ply

Depending on the application and the type of material being conveyed, nylon fabric breakers usually prove to be more effective in minimising the length of a rip compared to steel. The reason for this is that the nylon strands are able to stretch and pull together. As the trapped object is being pulled through the belt the strands of the breaker stretch and gather together into a bundle that can eventually become strong enough to stop the belt. Steel breakers are transversal steel wires held in position by longitudinal binder wires. As with fabric breaker plies, there are a wide range of strengths based on the size and pitch of the steel wires. Although having a steel wire breaker would logically seem to imply a much greater strength compared to a fabric ply breaker, this is not necessarily the case. Steel breaker wires do not stretch so they can not pull together to create a thicker barrier in the same way that fabric

plies can. However, the advantage of a steel breaker is that very sharp trapped objects (dolerite rock or slate for example) do not cut through the steel wires as easily compared to conventional nylon fabrics.

What creates these strengths is a very cleverly engineered ply construction that consists of extremely strong strands of polyester running longitudinally and heavy-duty nylon transverse strands held in position by a strong yarn. The strands are completely straight in both directions and are not interlocked in the conventional manner.

As with conventional breaker plies, the UsFlex fabric layer is embedded just above the steel cords. This also provides extremely effective impact protection of the steel cords because the fabric dissipates the peak point of impact energy over a much wider area.

POSITIONING OF THE BREAKER PLY

According to ISO 15236-1, a breaker ply should be positioned at a distance of between 1mm and 3mm from the longitudinal cords. The width of the breaker ply should be at least 10mm from the belt edge but no more than 100mm less than the width of the belt. On this basis, the breaker ply is regarded as part of the cover, which means that the cover thickness is measured from the cords.

However, if the ply is less than 1mm distance from the longitudinal cords then it is considered to be a weft transversal reinforcement and therefore part of the actual carcass.

This means that the cover thickness is measured outwards from the surface of the ply. It is generally advisable to position the breaker as close to the actual steel cords as possible because it maximises the amount of rubber that would have to be worn away by abrasion before the breaker ply is reached the surface of the cover.

When requesting quotations it is important to specify very clearly if the total thickness of the top cover should include the thickness of the breaker or not.

When there is a particularly high risk of accidental damage, it is possible to use a rip detection system as well as using breakers. The breakers are fitted in the top cover and the detection loops are fitted in the bottom cover below the cords. It is important to bear in mind that rip detection systems use electro-magnetic signals so they cannot function if a steel breaker ply is fitted.

This means that if a belt monitoring or rip detection system is going to be used then it is only possible to use fabric breakers.

CHOOSING THE BEST TYPE OF BREAKER

The primary factor to take into consideration when deciding on which type of breaker to use is the type of material being carried and how it is loaded on to the belt. Unless the materials being conveyed are particularly sharp (slate, granite, dolerite etc.) or there is a likelihood that it may contain sharp foreign objects (mining tools for example) then our recommendation is to use either conventional fabric breakers or the special UsFlex breaker ply.

This recommendation is based on extensive laboratory comparison tests and field experience, which show that transverse reinforcement with textile is more efficient in preventing longitudinal rip damage. Fabric breakers are also longer lasting than steel transverse reinforcement. If heavy materials such as large rocks are being dropped on to the belt surface then a full UsFlex breaker-ply layer is recommended.

When requesting a quotation, it is extremely important that potential suppliers are given a specification that is not subject to interpretation or misunderstanding.

This is essential not only in terms of the anticipated operational lifetime of the belt but also to help ensure that all potential suppliers are providing quotations based on exactly the same specifications. Because steelcord belts are usually only made to order it is advisable to order at least 50 meters of belt in addition to the required length so that it can be kept on site for emergency repairs.

Splicing

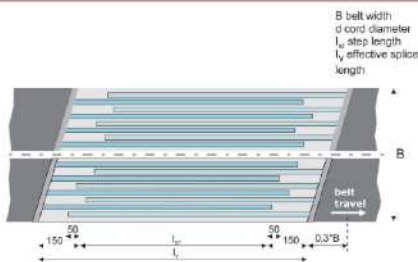
SINOCONVE Steelcord Conveyor

Belts are spliced by hot-vulcanization according to DIN standard. The splice length and number of steps depend on the belt rating. The standard splice is diagonal (approx. 17°), but rectangular splices are also possible. The pictures below illustrate the splice designs.

Particular attention must be paid to the splicing procedure. The method, conditions and the splicing kit have a great influence on the quality of the splice. For exact splicing instructions please contact SINOCONVE. It is strongly recommended to use splicing materials approved by the supplier.

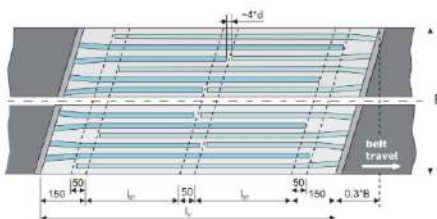
One-Step-Splice

Belt Type	Steps x step length l_{sr} in mm	Splice length l_s in mm
St 500	1 x 600	800
St 630	1 x 600	800
St 800	1 x 600	800
St 1000	1 x 600	800

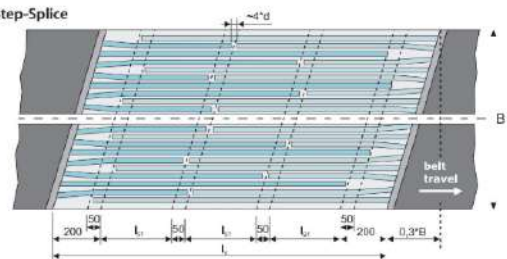


Two-Step-Splice

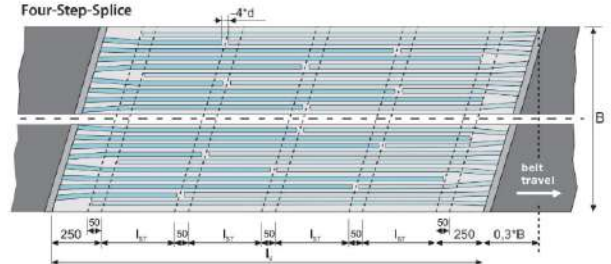
Belt Type	Steps x step length l_{sr} in mm	Splice length l_s in mm
St 1250	2 x 350	1050
St 1400	2 x 400	1150
St 1600	2 x 400	1150
St 1800	2 x 400	1150
St 2000	2 x 400	1150
St 2250	2 x 500	1350
St 2500	2 x 600	1550
St 2800	2 x 650	1650
St 3150	2 x 650	1650



Three-Step-Splice



Four-Step-Splice



Three-Step-Splice

Belt Type	Steps x step length l_{sr} in mm	Splice length l_s
St 3500	3 x 650	2450
St 4000	3 x 750	2750
St 4500	3 x 800	2900

Four-Step-Splice

Belt Type	Steps x step length l_{sr} in mm	Splice length l_s
St 5000	4 x 900	4750
St 5400	4 x 1000	4650
St 5800	4 x 1100	5050
St 6300	4 x 1200	5450
St 6700	4 x 1300	5850



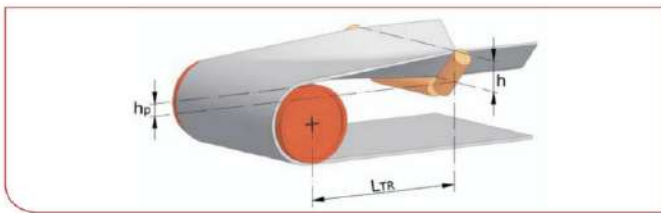
Transition Lengths

A reliable way to calculate minimum transition lengths (distance between tail/discharge pulley and first/last trough-hing idler) is to use the factors given in the table. The factors depend on the troughing angle and arrangement and must be multiplied by the belt width.

If the pulley is elevated with respect to the level of the center idler roll, the transition length can be decreased. Usually the elevation (h_p) is one third of the total troughing height (h).

(B = belt width)

Troughing angle	Minimum transition distance L_{tr}	
	$h_p=0$	$h_p=h/3$
20°	$1.5 \times B$	$1.0 \times B$
25°	$1.8 \times B$	$1.2 \times B$
30°	$2.2 \times B$	$1.5 \times B$
35°	$2.5 \times B$	$1.6 \times B$
45°	$3.0 \times B$	$2.0 \times B$

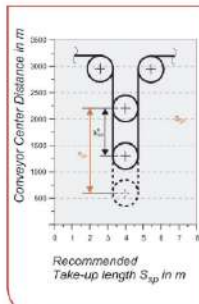


Travel of take-up pulley

Low elongation is one of the characteristics of SINOCONVE Steelcord Conveyor Belts. The travel of the take-up pulley is significantly less than for textile belts. When designing the take-up system, the take-up length S_{up} should not only consider travel of the take-up pulley S'_{up} due to the elastic elongation of the reinforcement. It is recommended to add also some surplus length for emergency splices, to compensate for

measuring errors or to correct the belt sag. When fitting the belt, make sure, that the take-up pulley has sufficient space in both directions to compensate maximum elongation when starting the conveyor under full load.

See the illustration for the recommended minimum take up length S_{up} .



QUALITY CONTROL

Step 1: Raw material inspection



Rubber compound test

Moving die rheometer analyzes the characteristics of Rubber Compound



EP fabric test

Tensile strength test and elongation test when broken.

Step 2: Advance equipment to guarantee quality from the details



Calender process

4 rollers calender machine is able to stick rubber on both side of the fabric in one time. This can reduce the strength loss of the rubber during the process.



Forming process

The forming tension is even between different plies, which can avoid wrinkled fabric and guarantee an even thickness of belt core.

QUALITY CONTROL

Step 3:Finished belts inspection

SINOCONVE BELT have an inspection team to inspect the surface and dimension of each conveyor belt. If any problem on belt surface, we will repair at once in our factory.



Length and surface inspection



Thickness inspection



Width inspection

Step 4: Technical data test

SINOCONVE BELT have a lab to test all raw materials and rubber compound. For each roll of conveyor belt, the lab will test the following features, and issue Quality Test Report.



Abrasion test



Adhesion strength test



Tensile strength test

TRANSPORT PACKAGE



QUCUSTOMER USE SITE



United Arab Emirates Quarry



Oman mines



Indonesia paper mill



Colombia Repairer



saudi arabia quarry



pakistan warehouse

